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# Nano positioning control for dual stage using minimum order observer †

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**Abstract**

A nano positioning control is developed using the ultra-precision positioning apparatus such as actuator, sensor, guide, power transmission element with an appropriate control method. Using established procedures, a single plane X-Y stage with ultra-precision positioning is manufactured. A global stage for materialization with robust system is combined by using an AC servo motor with a ball screw and rolling guide. An ultra-precision positioning system is developed using a micro stage with an elastic hinge and piezo element. Global and micro servos for positioning with nanometer accuracy are controlled simultaneously using an incremental encoder and a laser interferometer to measure displacement. Using established procedures, an ultra-precision positioning system (100 mm stroke and ±10 nm positioning accuracy) with a single plane X-Y stage is fabricated. Its performance is evaluated through simulation using Matlab. After analyzing previous control algorithms and adapting modern control theory, a dual servo algorithm is developed for a minimum order observer to secure the stability and priority on the controller. The simulations and experiments on the ultra precision positioning and the stability of the ultra-precision positioning system with single plane X-Y stage and the priority of the control algorithm are secured by using Matlab with Simulink and ControlDesk made in dSPACE.

*Keywords*: Nano; Positioning; Ultra-precision; Stage; Actuator; Sensor

## 1. Introduction

Nanotechnology can be applied to most industrial fields, including electronics, information and communication, mechanics, chemistry, bioengineering, and energy, and it exhibits a remarkable potential to change human civilization [1]. Ultra-precision positioning is a very important technology in the development of nano systems, and it encompasses a host of technologies, including mechanics, electronics, optics, control, design, and processing. With ultra-precision positioning, scanning tunneling microscopy (STM), or atomic force microscopy (AFM), measurements and manipulations at the single-atom level and operations in a small area of few square microns are possible. Currently, in the fields of ultra-micro processing, ultra-precision measurement, semiconductor wafers, optical communication, and opto-magnetic memory, the demand is for ultra-precision positioning allowing long strokes of hundreds of millimeters while maintaining nm-level precision. The basic technology to satisfy this demand is urgently needed [1-4].

Recent research into nano positioning concentrated on nano mechanisms and control methods. For example, the main emphasis of Lee's research [5] is the development of a control model for a dual servo stage, with the aim of achieving an overall positioning reproducibility of 10 nm for alignment of 200 mm diameter wafers. In a previous study [6], the focus was on developing an ultra-precision cutting unit (UPCU) that incorporated 3-axis control using piezo-electric actuators and on enhancing the precision of the current lathe used for ductile mode machining of hardened-brittle material, where the machining was based on the single crystal diamond. The machining data presented by H. S. Kim et al. show that their proposed approach allows the fabrication of aluminum mirrors 620 mm in diameter to a form accuracy of 0.7 µm (peak-to-valley error) [7]. In addition to this, much research has been done on nano positioning mechanisms and control methods to realize nano order control [8-14].

Therefore, in response to the demands in the field, a method is presented to produce an ultra-precision positioning machine that combines a micro and a global stage. In addition, a method is proposed for designing an optimum controller that uses a minimum order observer with a dual servo control scheme, and to which a modern control technique is applied.

## 2. Dynamic modeling

The ultra-precision positioning device proposed in this paper consists of a global stage onto which a micro stage is fixed. The global stage can be constructed in many ways; for this

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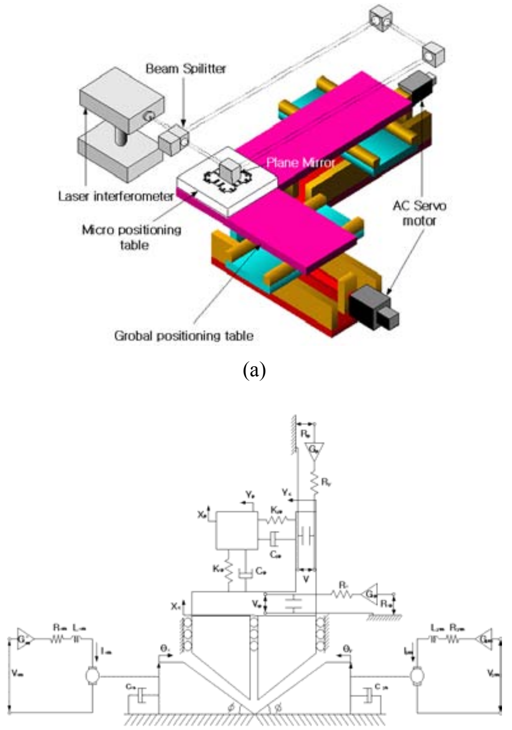



(a)

(b)

Fig. 1. Schematic of ultra-precision single plane X-Y stage: (a) system configuration; (b) dynamic modeling.

research the plane X-Y delta stage scheme is used, which uses a lead screw tool consisting of a ball screw and a double nut. This stage is driven by an AC servo motor (Mitsubishi MFS-23) and a drive. Motor revolutions are converted into linear motion by the double-nut type ball screw (2 mm lead and 19 mm external diameter) to convey global tables. In this case, the maximum power is 200 W, the maximum transfer distance is 100 mm, and the maximum transfer speed is 100 mm/s. For the micro stage,

I used the elastic hinge scheme and drive the stage using a piezoelectric sensor (Thorlabs AE1010D16). The maximum driving range is ±10 ㎛, and the stage is driven within the range of about ±2 ㎛. Fig. 1 shows an ultra-precision X-Y planar stage. To model this stage, the motion equation of the transfer screw of the global stage is first calculated, then the motion equation of the table and the circuit equation of the motor is calculated to find the transfer function by deriving the state equation.

The motion equation was derived by analyzing the relation between the micro stage displacement and the piezoelectric sensor and by measuring input voltage and the voltage across the piezoelectric sensor. Using this process, the transfer function may be found by deriving a state equation for the micro stage.

## 3. Constructing servo system

For this research, the ultra-precision plane X-Y stage is assumed to be a double-input single-output system, and an optimum control system was examined using modern control theory.

Because system design according to classical control theory is based on trial and error, it does not generate an optimum control system. Conversely, system design based on modern control theory using the state-space scheme allows designers to design systems having a closed loop polarity (i.e., the desired characteristic equation) or an optimum control for a given performance index. Furthermore, modern control theory allows designers to include initial conditions in their design if required. However, using modern control theory to design via the state-space scheme requires mathematical representation of the system's dynamic characteristics. This is different from the classical scheme, in which mathematical representations are not required and it is possible to use inaccurate experimental frequency response curves to design [5].

By exploiting the advantages of modern control theory, and on the basis of global and micro servo modeling, the double-input single-output state equation for an ultra-precision plane X-Y stage (described below) was derived, and a minimum order observer state-space scheme was designed. By so doing, it is possible to represent tools with a single state equation and to design global and micro servos as an integral body. Based on the state Eq. (1), I design the type-1 optimum servo system. The six state variables used in this system can be measured using a laser interferometer, and the remaining state variables can be measured using the minimum order observer. On this theoretical basis, the system was simulated and a control system was designed in which neither the micro-servo's extension nor the power current was saturated.

$$\dot{x}_s = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & 0 \\ 0 & 0 & a_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & a_{55} & a_{56} \\ 0 & 0 & 0 & 0 & a_{65} & a_{66} \end{bmatrix} x_s + \begin{bmatrix} 0 & 0 \\ b_{21} & 0 \\ b_{31} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & b_{62} \end{bmatrix} \begin{bmatrix} V_p \\ V_m \end{bmatrix} \tag{1}$$

$$x_s = \begin{bmatrix} x & \dot{x} & v & x_c & \dot{x}_c & i_m \end{bmatrix}^T \tag{2}$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} x_s \tag{3}$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \dot{y} \tag{4}$$

$$a_{21} = -\frac{k_p}{m},\quad a_{22} = -\frac{C_p}{m},\quad a_{23} = \frac{k_p k_p}{m} - \frac{C_p k_p}{RC_m}$$

$$a_{24} = -\frac{k_p}{m},\quad a_{25} = \frac{C_p}{m},\quad a_{33} = -\frac{1}{R_c}\quad a_{55} = -\frac{C_s}{J + k^2 M},$$

$$a_{56} = -\frac{kk_2}{J + k^2 M},\quad a_{65} = -\frac{k_1}{L_m k},\quad a_{66} = -\frac{R_m}{L_m},$$

$$b_{21} = \frac{C_p K_p}{RC_m} G_m,\quad b_{31} = \frac{K_m}{R_c},\quad b_{62} = \frac{G_p}{L_m}$$

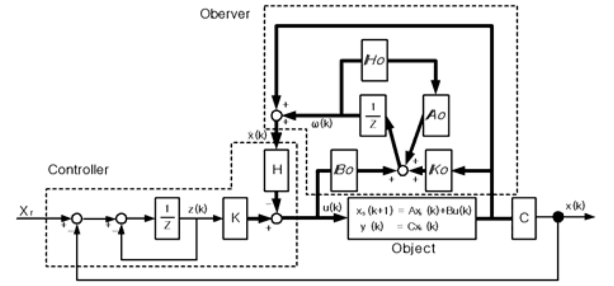



Fig. 2. Block diagram of the ultra-precise plane X-Y stage with a minimum order observer.

In actual systems some state variables can be directly measured, so that it is not necessary to estimate them. Given that n is the order of a state variable, the observer for estimating state variables with order less than n is called a minimum order observer. Assume a state vector x that is an n-vector and a power y that is an m-vector. Because m output variables are represented by a linear combination of n state variables, the minimum order observer has n–m orders.

The state variables in this system are of order six, and the measurable output vector is first order. As a result, the state variable x can be measured by laser interferometer, and the remaining state variables are estimated from the minimum order observer. Fig. 2 shows a minimum order observer applied to this system.

The state equation just derived of the ultra-precision, double-input single-output plane X-Y stage can be expressed in the form of a minimum order observer as a matrix, as shown below:

$$A_{aa} = [0] \qquad A_{bb} = [1 \quad 0 \quad 0 \quad 0]$$

$$A_{ba} = \begin{bmatrix} a_{21} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad A_{bb} = \begin{bmatrix} a_{22} & a_{22} & a_{24} & a_{25} & 0 \\ 0 & a_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & a_{55} & a_{56} \\ 0 & 0 & 0 & a_{65} & a_{66} \end{bmatrix}$$

$$B_a = [0 \quad 0] \qquad B_b = \begin{bmatrix} b_{21} & 0 \\ b_{31} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & b_{62} \end{bmatrix}. \tag{5}$$

Here, the equation of a measurable state variable is described in Eq. (6):

$$\dot{x}_a = A_{aa}x_a + A_{ab}x_b + B_a u \,. \tag{6}$$

Also, Eq. (6) was translated to derive the following Eq. (7):

$$\dot{x}_a - A_{aa}x_a - B_{au} = A_{ab} \cdot x_b \,. \tag{7}$$

The left term in Eq. (7) can be measured and may be regarded as an output equation. When designing a minimum order observer, the left term in Eq. (7) is the displacement of the system, and Eq. (7) gives the relationship between a measurable state variable and non-measurable variable. Here, $x_a$ is a state variable; the remaining state variables are $x$, $xb$, $\dot{x}$, $\upsilon$, $x_c$, $\dot{x}_c$ $i_m$ . The variables that cannot be measured are related by

$$\dot{x}_b = A_{ba}x_a + B_b u \,. \tag{8}$$

Eq. (8) expresses a dynamic characteristic of a state variable that is not measured. The state equation for a minimum order observer is Eq. (8), and the output equation is Eq. (7). Therefore, the equation of a minimum order observer is

$$\dot{\tilde{x}}_b = (A_{bb} - k_e A_{ab})\tilde{x} + A_{ba}x_a + B_a u + k_e(\dot{x} - A_{aa}x_a - B_a u) \tag{9}$$

where $k_e$ is the gain matrix for an $[(n\text{-}1)\times 1]$ state observer. To estimate Eq. (9), the derivative of $x_a$ is required. However, this is not desirable, so Eq. (9) is converted to

$$\dot{x}_b - k_e\dot{x}_a = (A_{aa} - K_e A_{ab})\dot{x}_b + (A_{ab} - k_e A_{aa})\mathrm{y} + (B_b - k_e - B_a)u \tag{10}$$

using

$$x_b - k_e \mathrm{y} = x_b - k_e x_a = \eta$$
$$\tilde{x}_b - k_e y = \tilde{x}_b - k_e x_a = \tilde{\eta} \tag{11}$$
$$\tilde{\eta} = (A_{bb} - k_e A_{ab})\tilde{\eta} + [(A_{bb} - K_e A_{ab})k_e + A_{ba} - k_e A_{aa}]\mathrm{y} + (B_a - k_e B_a)u$$
$$+[(A_{bb} - K_e A_{ab})k_e + A_{ba} - k_e A_{aa}]y + (B_a - k_e B_a)u \,. \tag{12}$$

Eqs. (11) and (12) are used to define a minimum order observer.

To derive the fifth-order equation of a minimum order observer, Eq. (7) is used in Eq. (9), resulting in

$$\tilde{\eta} = (a_{aa} - k_e A_{ab})\tilde{\eta} + A_{ba}x_a + B_a u + k_e A_{ab}x_b \,. \tag{13}$$

Subtracting Eq. (13) from Eq. (6) gives

$$\dot{x} - \dot{\tilde{x}}_b = (a_{bb} - k_e A_{ab})(x_b - \tilde{x}_b) \tag{14}$$

using

$$e = x_b - \tilde{x} = \eta - \tilde{\eta} \tag{15}$$

where e is the equation of order n –1, and Eq. (14) gives

$$\dot{e} = (A_{aa} - K_e A_{ab})e \,. \tag{16}$$

Thus, the fifth-order equation for a minimum order observer is given by Eq. (16). A characteristic equation of a minimum order observer is

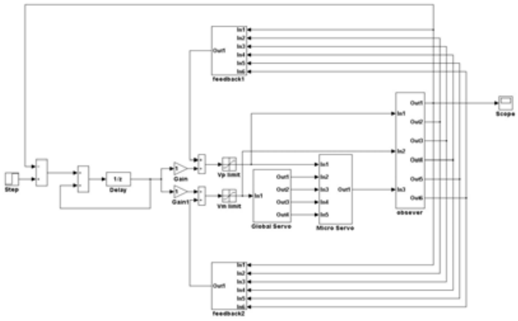


Fig. 3. SIMULINK for a dual servo for an ultra-precision single plane X-Y stage.

$$
\begin{aligned}
&\left|SI - A_{bb} + k_e A_{ab}\right| \\
&= (S-\mu_1)(S-\mu_2)(S-\mu_3)(S-\mu_4)(S-\mu_5) \\
&= S^5 + a_1S^4 + a_2S^3 + a_3S^2 + a_5
\end{aligned}
\tag{17}
$$

where $\mu_1$, $\mu_2$, $\mu_3$, $\mu_4$, $\mu_5$ are the desired characteristic values of a minimum order observer.

The gain matrix $k_e$ of the observer is obtained after selecting the characteristic values. Using the Ackeomann formula [15], the gain matrix of an observer is

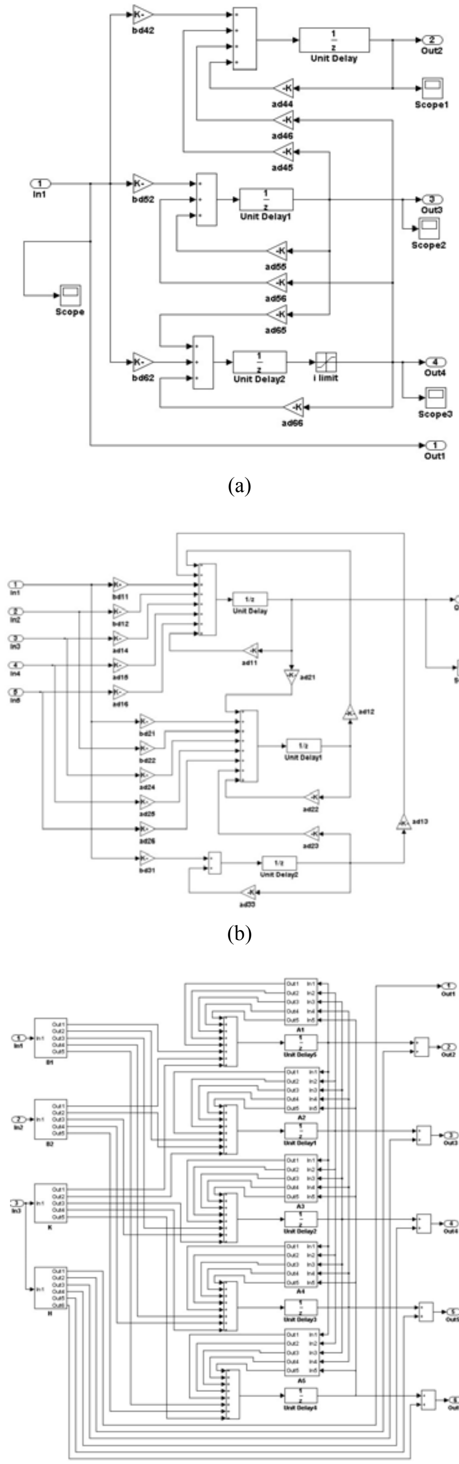
$$
k_e = \Phi\cdot(A_{bb})\begin{bmatrix} A_{ab} & \\ A_{ab} & A_{bb} \\ A_{ab} & A_{bb}{}^2 \\ A_{ab} & A_{bb}{}^3 \\ A_{ab} & A_{bb}{}^4 \end{bmatrix}^{-1}\begin{bmatrix}0\\0\\0\\0\\0\\1\end{bmatrix}
\tag{18}
$$

$$
\Phi(A_{bb}) = A_{bb} + a_1 A_{BB}{}^4 + a_3 A_{bb}{}^2 + a_1 I \ .
$$

Using this process, the gain matrix $k_e$ of an observer and the matrix $/A_0$, $/B_0$, $/H_0$, $/K_0$ of the observer may be obtained. Each equation matrix is given below:

$$
\begin{aligned}
&/A_0 = A_{bb} - K_e A_{ab}(5\times 5) \\
&/B_0 = B_b - K_{eB_a}(5\times 5) \\
&/K_0 = A_{ab-}k_e A_{aa}(5\times 1) \\
&/H_0 = \left[-\frac{1}{k_e}\right](6\times 1)\ .
\end{aligned}
\tag{19}
$$

In testing the ultra-precision positioning of the system, simulation and real-time control was performed with MATLAB SIMULINK. The experiment consisted of a single servo experiment and a dual servo control test. In the single servo control test, simulation and real-time control were performed using a PID controller for both the global and the micro servo, respectively. For the dual servo control test, I applied the minimum order observer using modern control theory to compare the results of simulation and real-time control. For dual servo simulation, the minimum order observer under modern control theory was applied to the state equation of a double-input single-output plane stage (derived above) to construct a type-1 optimal servo system, and the system was simulated using MATLAB SIMULINK.



(a)

(b)

(c)

Fig. 4. Sub-SIMULINK of dual servo for ultra-precision single plane X-Y stage: (a) SIMULINK for global servo; (b) SIMULINK for micro servo; (c) SIMULINK minimum order observer.

Fig. 3 shows a dual servo SIMULINK, and Fig. 4 shows

Table 1. Step response of the global servo used for simulation.

| Definition | Value |
|---|---|
| Steady-state error | 0 |
| Maximum overshot [%] | 0.1(10%) |
| Rise time [s] | 0.047 |
| Setting time [s] | 0.062 |

Table 2. Step response function for micro servo used for simulation.

| Definition | Value |
|---|---|
| Steady state error | 0 |
| Maximum overshot [%] | 0.1(10%) |
| Rise time [s] | 0.005 |
| Setting time [s] | 0.0075 |

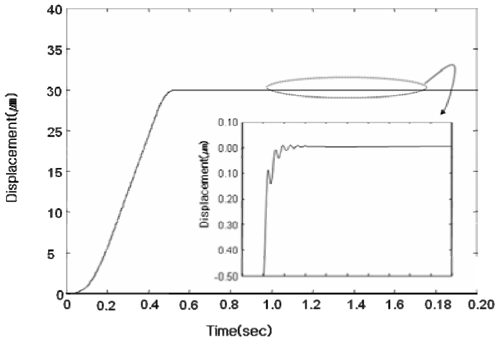



Fig. 5. Step response of the dual servo used for simulation.

SIMULINK of a minimum order observer. Fig. 5 shows the dual stage displacements including a global and a micro servo, and the simulated displacement of the global stage according to the SIMULINK dual servo simulation. Fig. 5 shows the displacement of a micro stage, where we see that the system is driven while the global stage error is corrected by the micro stage, indicating that the correct gain control is operating. Tables 1 and 2 show simulated control performance of a global and a micro servo.

A PID controller was installed on a global and a micro servo to perform positioning tests, and satisfactory results were obtained for each tool. These experimental results demonstrate the stability of the global servo and the micro servo. By applying the minimum order observer based on modern control theory for ultra-precision position control, real-time control of a dual servo was carried out. Fig. 6 shows a photograph of this system, and the real-time control RTI SIMULINK of a dual servo is shown in Fig. 7. Initialization is performed at program startup. Fig. 7 shows the RTI SIMULINK of the X-axis (Y-axis has the same structure).

Micro servo step response of Fig. 7 showed the same result

Fig. 6. Photo of ultra-precision single plane X-Y stage.

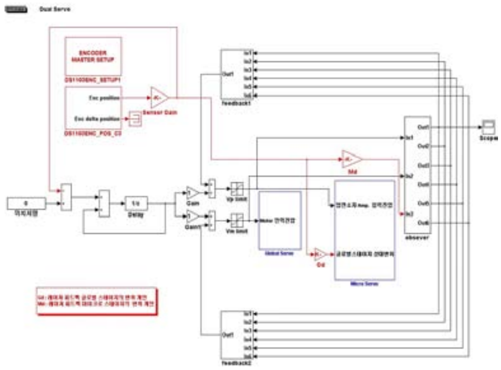
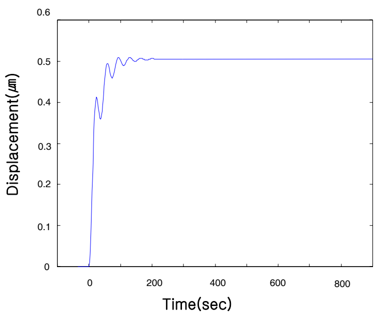
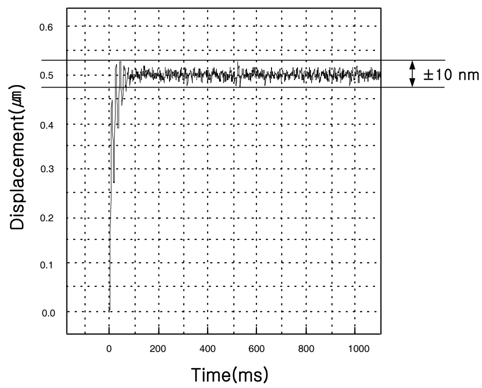


Fig. 7. RTI Simulink of dual servo.




(a) Simulation




(b) Experiment

Fig. 8. The simulation results and the ultra-precision positioning experiment result.

as compared simulation of Fig. 5 to experiment result of Fig. 8, and resolution of 10 nm can be found.

Figs. 9 and 10 show a step experiment and a resolution ex-

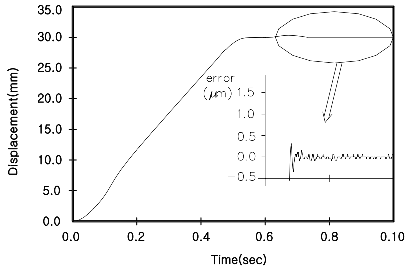



Fig. 9. Step experiment of X-Y axis dual positioning control.

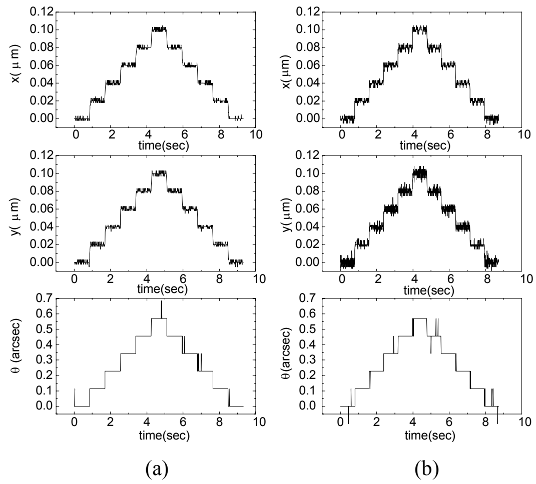



Fig. 10. Resolution experiment of micro positioning control on a vibration-isolation table: (a) with the table on; (b) with the table off.

periment of a dual servo, respectively. To evaluate the ultra-precision positioning capability of the system, two positioning experiments were conducted, disregarding the error value measured earlier. In the first experiment, a 20 nm step pulse was used as input to generate a linear displacement of up to 100 nm on a vibration-isolation table (table on), where the external vibration was not notified to system, which was followed by 20-nm steps in the opposite direction to get back to the origin. Each displacement was configured to move 0.57 arcsec in steps of 0.114 arcsec. The data shown in Fig. 10(a) confirm that no difficulty occurred during the detailed driving of the 20-nm steps and 0.114 arcsec steps. Furthermore, the linear displacement was10 nm and the positioning was 0.114 arcsec.

In the second experiment, a 20 nm step pulse input was used to generate a linear displacement of up to 100 nm on the vibration-isolation table (table off). All the other aspects of the experiment were the same as described for the first experiment. The data shown in Fig. 10(b) indicate that the overall performance of the detailed driving was lower than that obtained using the vibration-isolation table.

## 4. Conclusions

The research on designing an optimum controller for an ultra-precision plane X-Y stage yielded the following results:

(1) Using a Matlab simulation, the performance of the control system for a servo tool that uses an ultra-precision plane X-Y stage positioning system was evaluated. A dual servo control algorithm was devised using a minimum order observer to which modern control theory was applied by comparing and analyzing the conventional control algorithm. The controller was verified as having good stability and performance.

(2) The simulation results and the ultra-precision positioning experiment result were compared and analyzed through a real-time ultra-precision positioning experiment using Matlab Simulink and ControlDesk from dSPACE. The position control algorithm and the plane X-Y stage ultra-precision positioning system were verified as having good performance and stability.

(3) The positioning test of the system indicates that the resolution for linear translation and rotation is 10 nm and 0.144 arcsec, respectively.

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